

Ch10_Perspectives for decision-making using (generative) AI

Introduction

社会趋势：数据驱动决策 Data– driven decision– making

- Analyzing, interpreting, and presenting data meaningfully

技术浪潮：生成式人工智能（Generative AI）

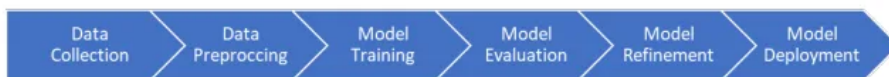
- Creating new data or content from existing data

反馈加速器：自然语言生成（NLG）

- Generating natural language texts from non– textual inputs

Generative AI System Development

1. **Data collection** – various sources
2. **Data preprocessing** – Cleaning, transforming, and formatting data (Remove noise, outliers, duplicates...)
3. **Model training** – Using machine learning algorithms to learn patterns (supervised, unsupervised, semi–supervised learning).
4. **Model evaluation**– Assessing performance and quality of the model (quantitative and qualitative evaluation methods).
5. **Model refinement** – Improving model parameters or components 优化模型参数或组件
6. **Model deployment** – Making the model available for use (deploy via APIs, web apps; ensure scalability, reliability, security). 通过 API、Web 应用部署；确保可扩展性、可靠性与安全性



Types and applications of Generative AI

- **Text generation:**
 - Natural language processing and generation
- **Image generation:**
 - Creating realistic or artistic images
- **Audio generation:**
 - Producing speech, music, or sound effects
- **Video generation:**
 - Creating animations, simulations, or manipulations

Agents in Generative AI

- Interactive systems using language understanding
- Capabilities:
 - Make plans to solve problems or tasks
 - Execute actions using tools or APIs
 - Interact with users for feedback and refinement
- Example workflow:
 - Understand user query
 - Create a plan of action
 - Execute actions using tools/APIs
 - Generate response
 - Get user feedback and refine

- **Interactive systems** using language understanding – 基于语言理解的交互式系统

这些代理（Agents）不仅能“听懂”用户的自然语言，还能主动规划任务、调用工具、生成响应，并根据用户反馈持续优化。

- **核心能力 Capabilities**

Make plans to solve problems or tasks	制定解决问题或任务的行动计划
Execute actions using tools or APIs	使用工具或 API 执行具体操作
Interact with users for feedback and refinement	与用户互动以获取反馈并持续优化

这些能力构成了智能代理的闭环执行逻辑：

- **计划（Plan）**：理解目标，制定步骤
- **执行（Act）**：调用工具完成任务
- **反馈（Refine）**：根据用户反馈调整策略

a	Understand user query	理解用户请求
b	Create a plan of action	制定行动计划
c	Execute actions using tools/APIs	使用工具或 API 执行操作

d	Generate response	生成响应内容
e	Get user feedback and refine	获取用户反馈并进行优化

- 意义：
 - From static generation to dynamic agency
 - Feedback– driven refinement
 - Tool– integrated execution
 - Multi– step task automation 多步骤自动化

Opportunities and Risks with Generative AI

Opportunity	Risk
Enhancing creativity Generates novel ideas and solutions	Reducing human creativity Over–reliance may suppress original thought
Improving personalization Tailors outputs to individual preferences	Privacy concerns Misuse of personal data in training
Increasing automation Streamlines repetitive tasks	Errors or inconsistencies Automated systems may propagate mistakes Automation boosts efficiency, yet errors can scale rapidly
Expanding scalability Handles large volumes of data and tasks 扩展可扩展性：处理大量数据与任务	Decision overload Too many options may paralyze decision–makers 信息过载：选项太多导致决策瘫痪
Boosting accuracy Provides precise, data–driven insights	Biased or inaccurate outputs Models may amplify data bias; Accuracy depends on data quality; poor data leads to flawed results

Promoting diversity Generates varied perspectives and options	Reducing diversity Outputs may homogenize due to training limits; AI can broaden perspectives, but may also converge toward sameness
Supporting ethicality Embeds ethical guidelines into decisions	Ethical challenges Potential for harmful or inappropriate content

Evaluation Generative AI model Accuracy

- Quantitative evaluation:
 - Numerical scores (e.g., IS, FID, PR, perplexity)
 - Metrics for specific tasks (e.g., BLEU, ROUGE for text)
- IS (Inception Score)：用于图像生成，衡量生成图像的清晰度与多样性
- FID (Fréchet Inception Distance)：衡量生成图像与真实图像之间的距离
- PR (Precision & Recall)：衡量生成内容的准确性与覆盖率
- Perplexity (困惑度)：用于语言模型，衡量模型对文本的预测能力，数值越低越好
- BLEU / ROUGE：用于文本生成，衡量生成文本与参考文本的相似度（BLEU偏向精确度，ROUGE偏向召回率）
- Qualitative evaluation:
 - Human judgments on realism, coherence, relevance
 - User feedback and studies

定性评估依赖人类感知与主观判断，适合评估模型在实际应用中的表现。

- Realism (真实感)：生成内容是否像人类创作的
- Coherence (连贯性)：内容是否逻辑通顺、上下文一致
- Relevance (相关性)：是否与用户需求或输入主题相关
- User Feedback (用户反馈)：通过问卷、访谈、使用日志等方式收集用户体验
- User Studies (用户研究)：设计实验场景，观察用户与模型交互的效果

Responsible and Ethical Approach

Challenges in Generative AI

1. **Data Quality** 数据质量 – Ensuring accuracy, completeness, and relevance

生成式 AI 的输出质量高度依赖训练数据。如果数据不准确、缺失或与任务无关，模型可能生成错误或无意义的内容。

2. **Data Availability** 数据可用性 – Obtaining **sufficient data** for training and evaluation

数据不足也会导致过拟合或偏见。

3. **Data Ownership** 数据所有权 – Respecting rights and interests of data providers

使用第三方数据进行训练时，必须考虑版权、许可与使用范围。忽视数据所有权可能引发法律纠纷，尤其在商业化模型中更为敏感。

4. **Data Privacy** 数据隐私 – Protecting confidentiality of data subjects

5. **Model Robustness** – Handling changes and uncertainties in data or environment

面对新情况、异常输入或环境变化时仍能稳定运行。

6. **Model Accountability** 模型责任归属 – Assigning responsibilities for model outputs

当生成内容引发误导、歧视或伤害时，谁来负责？是开发者、使用者还是平台？模型责任归属是 AI 治理中的核心议题，涉及伦理、法律与社会规范。

Best Practices for Responsible Generative AI Use

- **Understanding AI model limitations** 辨别信息 based on your critical thinking

- **Addressing biases:**

- Identifying and mitigating biases in data and models

- **Maintaining transparency:**

- Clearly communicating AI involvement and capabilities

- **Securing sensitive information:**

- Implementing strong data protection measures

- **Ensuring accountability and explainability:**

- Providing clear explanations for AI decisions

- **Continuous monitoring:**

- Regularly assessing model performance and impacts

- **Educating and empowering users:**

- Providing training on responsible AI use

- **Transparency** is about **openness**, making the system's architecture, training data, and usage visible to stakeholders.

- **Explainability** is about **understanding**, making individual decisions or predictions interpretable and justifiable.

Ethical Principles for Generative AI

- Informed consent:
 - Obtaining user agreement for data use
 - Aligned interests:
 - Ensuring AI goals match user/societal interests
 - Opt-in and easy exits:
 - Allowing users to control their participation
 - Conversational transparency:
 - Clearly identifying AI in interactions
 - Debiased and explainable AI:
 - Mitigating biases and providing explanations
 - AI training and development:
 - Ethical considerations in model creation
 - Health and well-being considerations:
 - Assessing impact on users and society
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- Health：身体健康、安全、无害
 - Well-being：心理状态、情绪体验、社会影响、生活质量
 - 健康与福祉考量：指 AI 系统在使用过程中是否对用户或社会造成身体、心理或社会层面的负面影响，是否有助于提升用户的生活质量与安全感。

应用举例：

- AI 聊天助手是否会误导心理脆弱用户？
- 推荐算法是否加剧焦虑、沉迷或信息茧房？
- 医疗 AI 是否真正提升诊断质量而非制造风险？

Consent	Informed agreement to data use	知情同意：用户明确了解用途后同意使用其数据
Opt-in	Voluntary enrollment or participation	主动加入：用户选择参与某项服务或功能

- Consent 是法律与伦理层面的授权，比如“我同意你使用我的数据用于训练 AI”。
- Opt-in 是产品设计层面的机制，比如“我勾选了这个功能”或“我点击了加入”。

Legal and Regulatory Considerations

- Data collection and sharing practices:
 - Compliance with data protection laws
- Privacy and security measures:
 - Implementing robust safeguards
- Third-party disclosure policies:
 - Managing data sharing with external entities
- Compliance with laws and regulations:
 - Adhering to GDPR, AI Act, and other relevant legislation

1. Data Collection and Sharing Practices

在收集和共享用户数据时，必须遵守相关法律 comply with relevant data protection law，如 GDPR（欧盟通用数据保护条例）、中国《个人信息保护法》等。这包括明确告知用户数据用途、获取同意、限制数据滥用等 This includes clearly informing users about how their data will be used, obtaining explicit consent, and implementing safeguards to prevent misuse.

2. Privacy and Security Measures

为了防止数据泄露、滥用或非法访问，组织需部署加密技术、访问控制、身份验证机制等安全措施。同时也要确保数据在传输和存储过程中保持机密性与完整性。

3. Third-Party Disclosure Policies 第三方数据披露政策

Managing data sharing with external entities	管理与外部机构的数据共享行为
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数据需要与第三方（如合作伙伴、服务商）共享时，必须制定清晰的政策，明确共享范围、目的、权限与责任，防止数据滥用或泄露。

4. Compliance with Laws and Regulations 遵守法律与法规

不仅要遵守数据保护法，还要关注 AI 相关法规，如欧盟《人工智能法案》（AI Act），确保模型开发、部署与使用过程合法合规，避免伦理与法律风险。

Data Collection & Sharing	数据收集与共享	合规采集与合理共享	防止非法使用数据
Privacy & Security	隐私与安全	加强数据保护机制	防止泄露与攻击
Third-Party Disclosure	第三方披露	管理外部数据共享	控制风险边界
Legal Compliance	法律合规	遵守数据与 AI 法规	确保合法运营与责任清晰